DP Biology – D1.1

DNA replication

- How do cells make new DNA?
- What biotechnology has understanding DNA replication made possible?

SL statements D1.1.1–5. Textbook pp. 602–610.

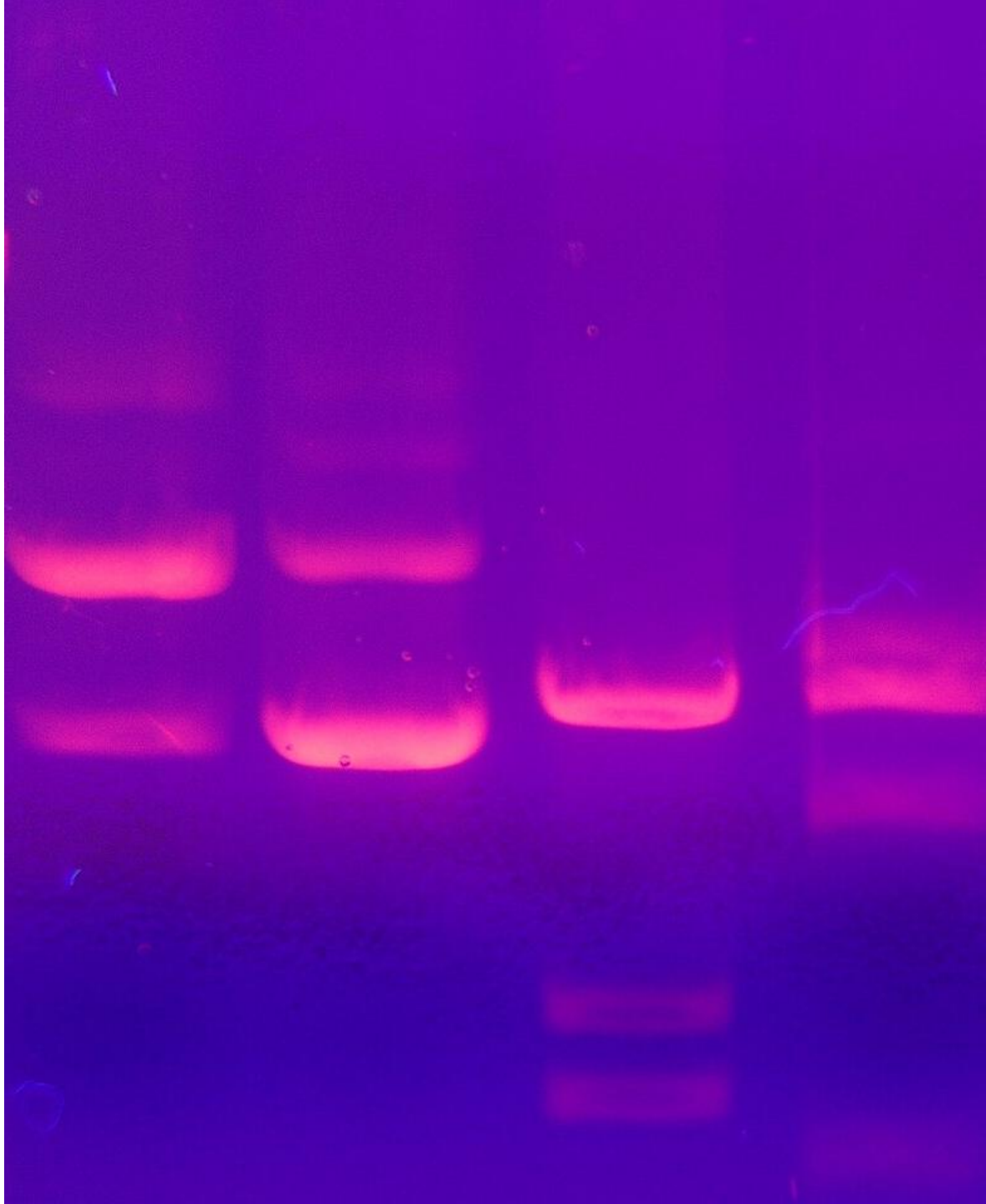


Image: DNA fragments on an electrophoresis gel. Mnolf, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	D1.1.1–3	Explain why DNA is replicated, why replication is semi-conservative and accurate, and what helicase and DNA polymerase do
2	D1.1.4	Explain how PCR copies DNA and how gel electrophoresis separates fragments
3	D1.1.5	Use a DNA profile to reach a conclusion in forensic and paternity cases

Why cells copy their DNA

DNA replication makes exact copies of DNA, keeping the base sequence the same. It happens before a cell divides.

Reproduction

Offspring need copies of the parents' base sequences.

Growth

Every new cell needs a full set of DNA.

Tissue replacement

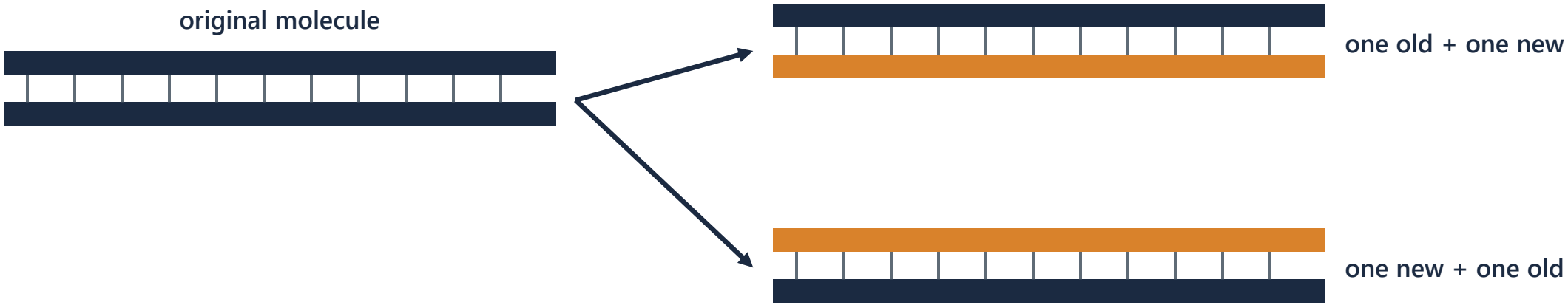
Lost cells, such as skin, are replaced with identical genes.

Common mistake

Replication is not cell division. DNA is copied first (in interphase); the cell divides later.

Replication is semi-conservative

D1.1.2



Navy: old strands. Orange: new strands. Drawn, not to scale.

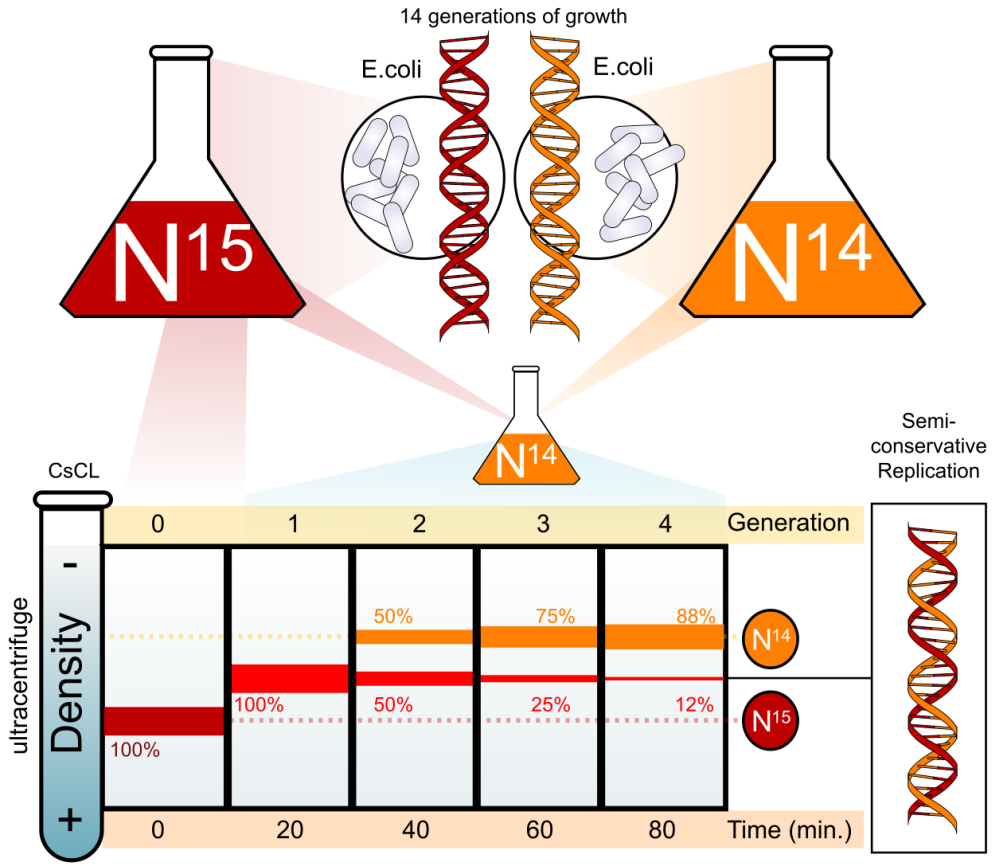
The strands separate and **each acts as a template**. Every new molecule has **one old strand and one new strand**: semi-conservative.

Why it matters

Each copy keeps a strand checked against the original, so the sequence is conserved.

How the model was tested

D1.1.2



The Meselson-Stahl experiment. LadyofHats, public domain.

Bacteria were grown on **heavy** nitrogen, then moved to **light** nitrogen. DNA was spun to separate it by density.

After one generation all DNA was **intermediate**; after two, half intermediate and half light. Only semi-conservative replication predicts this.

Think

Predict the bands after one generation if replication were conservative.

Base pairing makes copying accurate

D1.1.2

Strand	Base sequence
Template	A T G A C C G C T
New strand	T A C T G G C G A

A pairs only with **T**; **C** pairs only with **G**. Only complementary bases form hydrogen bonds in the right position.

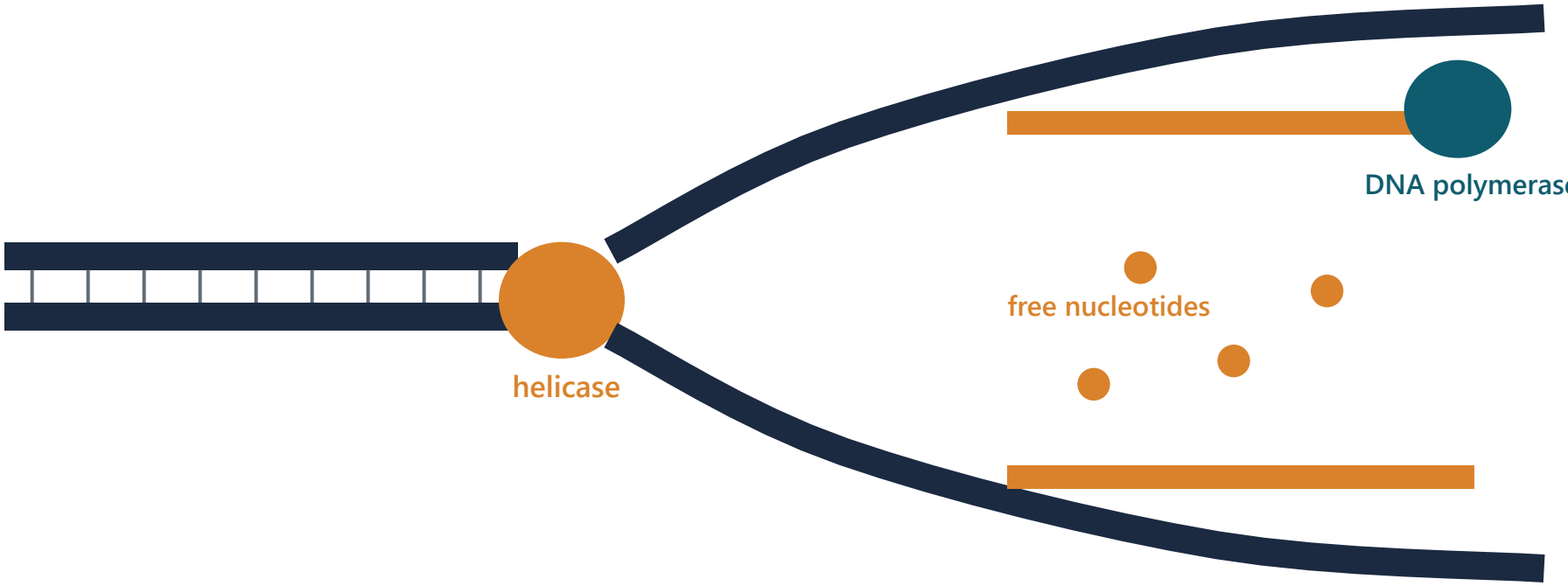
So the template decides which nucleotide joins next, and both copies match the original.

Key idea

Mistakes are rare, but when they happen they are the source of mutations (D1.3).

Helicase and DNA polymerase

D1.1.3



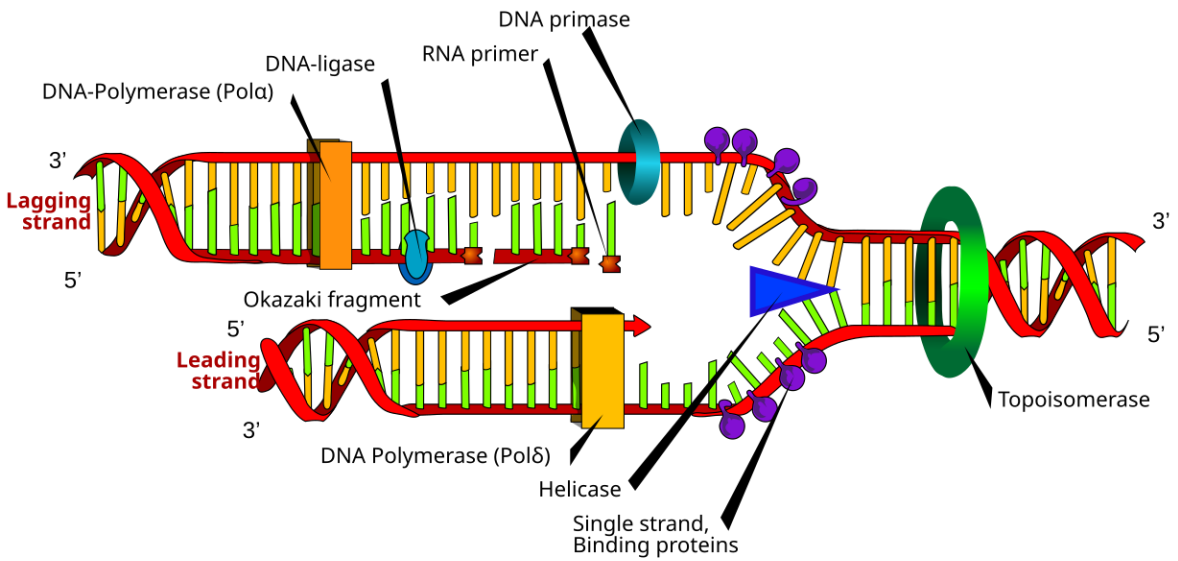
Helicase unwinds the helix and breaks hydrogen bonds between bases.

DNA polymerase links free nucleotides into a new strand, matched to the template.

Drawn replication fork, not to scale. Navy: template. Orange: new strand.

The two enzymes compared

D1.1.3



Enzyme	Role at SL
Helicase	Unwinds; breaks hydrogen bonds between bases
DNA polymerase	Joins nucleotides into a new strand on the template

A replication fork. LadyofHats, public domain. HL covers the other enzymes.

Bonds

Helicase breaks **hydrogen bonds** between strands. Polymerase makes **covalent bonds** along the new backbone.

Practice: copy and predict

D1.1.2–3

Strand	Sequence
Template 1	T A C G G A T C
New strand 1	?
Template 2	G G C A T T A C
New strand 2	?

1. Write each new strand.
2. After two rounds, what fraction of DNA molecules contain an original strand?
3. Which enzyme would be blocked by a drug that stops strands separating?

Check-in: class 1

D1.1.1–3

1. Give three reasons why DNA is replicated.
2. Explain the term semi-conservative.
3. Explain how base pairing makes replication accurate.
4. State the role of helicase.
5. State the role of DNA polymerase.

How to use this

Answer without your notes, then check against slides 3–9.

PCR copies a chosen section of DNA

D1.1.4



A thermal cycler. Juan Carlos Fonseca Mata, CC BY-SA 4.0.

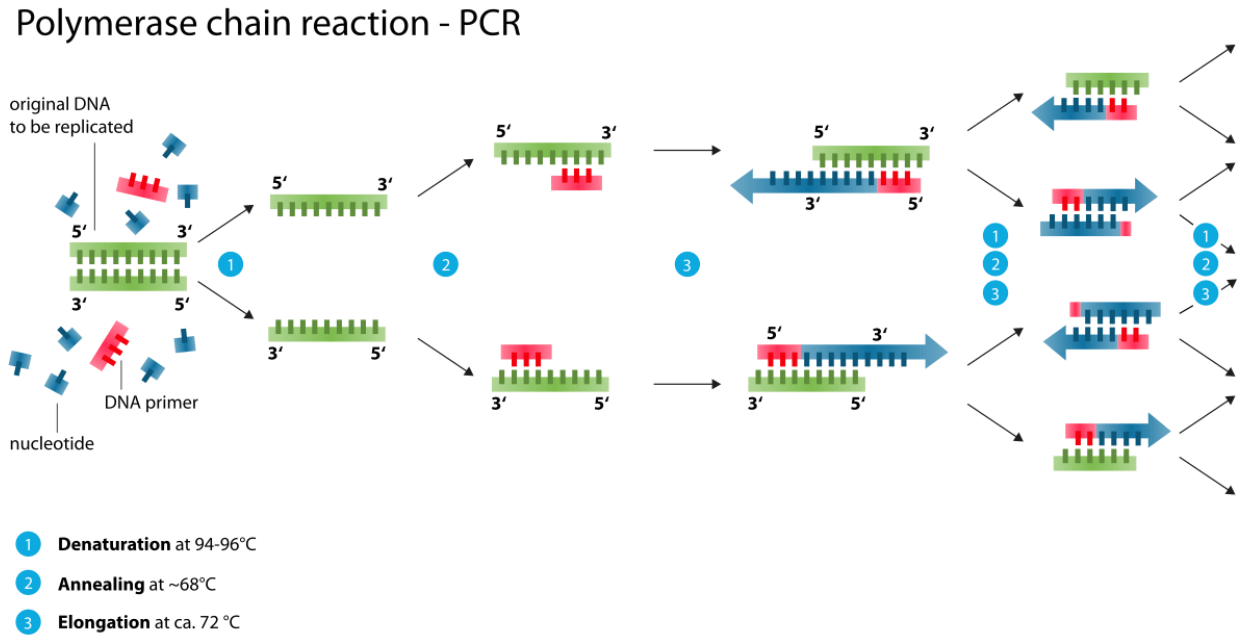
Ingredient	Why it is needed
DNA sample	Holds the section to be copied
Primers	Short DNA strands that mark the start of the section
Nucleotides	Building blocks for new strands
Taq polymerase	Builds new strands; survives high temperatures

Key idea

The **polymerase chain reaction** amplifies a tiny sample. A thermal cycler changes the temperature in each cycle.

One cycle of PCR

Denaturation 95 °C: strands separate → **Annealing** about 55 °C: primers attach → **Extension** 72 °C: Taq builds new strands

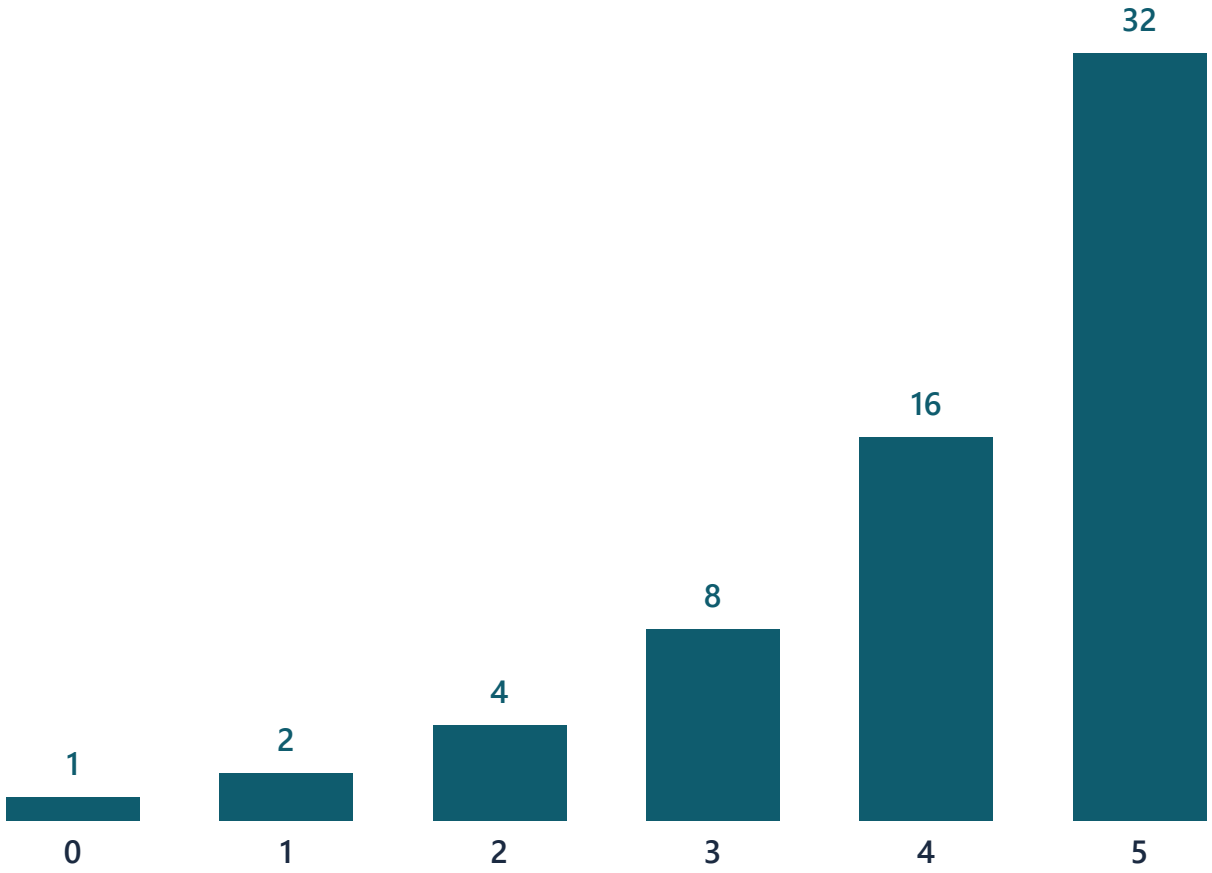


Steps in a PCR cycle. Enzoklop, CC BY-SA 3.0.

Heating breaks only the **hydrogen bonds**; the backbones stay intact.

The cycle repeats 25–35 times.

Each cycle doubles the DNA



Cycle number. Copies from one molecule above each bar.

1 → 2 → 4 → 8 ... The number **doubles** every cycle: after 10 cycles, 1024 copies.

After 30 cycles: over a billion copies, in about two hours.

Why Taq polymerase is used

D1.1.4



A hot spring in Yellowstone. Jim Peaco, National Park Service, public domain.

Taq comes from **Thermus aquaticus**, a bacterium that lives in hot springs.

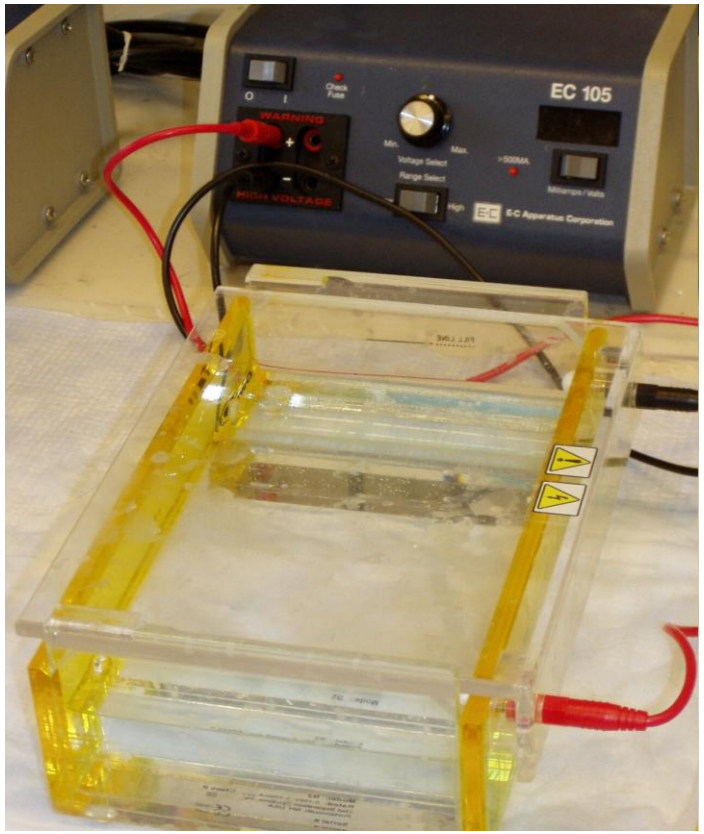
It is **not denatured at 95 °C**, so it survives every cycle. Most polymerases would be destroyed in the first.

Link

An adaptation to life in hot water became a key tool in biotechnology.

Gel electrophoresis separates fragments

D1.1.4



Loading and running a gel. Jeffrey M. Vinocur, CC BY 2.5.

DNA samples go into **wells**; a current runs across the gel. DNA is **negatively charged** (phosphate groups), so it moves towards the **positive** electrode.

The gel acts as a sieve: **small fragments travel furthest**. A **ladder** of known lengths is run for comparison.

Common mistake

All fragments carry negative charge. Charge makes them move; length separates them.

Reading a gel



wells at top; positive electrode at the bottom

Each band holds many fragments of the same length. Bands near the wells are the **longest**.

Think

Which sample has the shortest fragment? Which shares a band with each of the other two?

Check-in: class 2

D1.1.4

1. List the ingredients of a PCR reaction.
2. Outline the three steps of a PCR cycle with temperatures.
3. Calculate the copies after 10 cycles from one molecule.
4. Explain why Taq polymerase is used.
5. Explain why DNA fragments separate in a gel.

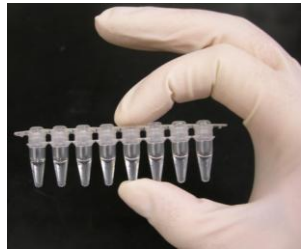
How to use this

Answer without your notes, then check against slides 11–16.

What PCR and electrophoresis are used for

D1.1.5

Application	How the tools help
Forensics	Tiny crime-scene samples amplified and compared with suspects
Paternity	Child's fragments compared with mother and possible father
Diagnosing infection	PCR detects a pathogen's genes (RNA first copied into DNA)
Research	Enough copies of a gene to study or sequence

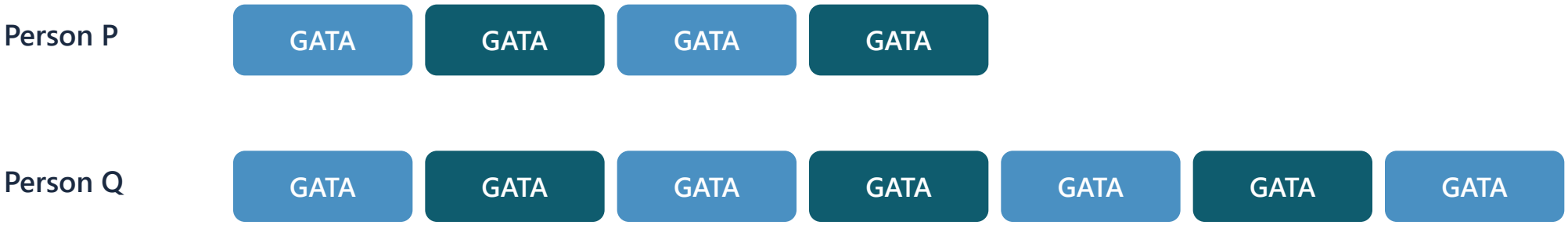


PCR tubes hold the tiny volumes used. Image: Madprime, CC0.

Why DNA differs between people

D1.1.5

Some non-coding regions contain a short sequence repeated many times: a **short tandem repeat (STR)**. The **number of repeats** varies between people.

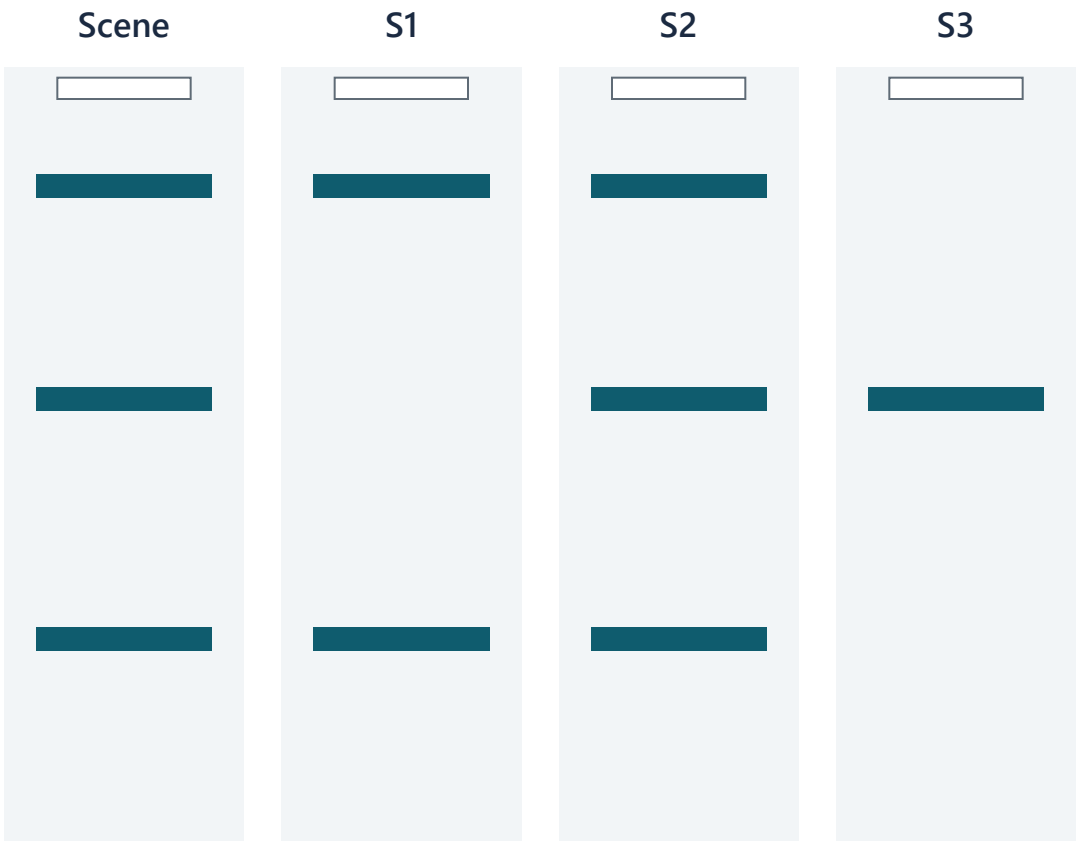


Key idea

PCR copies these loci; electrophoresis separates them by length. Each person has two copies of each locus, one from each parent. Comparing many loci makes a chance match between unrelated people very unlikely.

Forensic investigation

D1.1.5



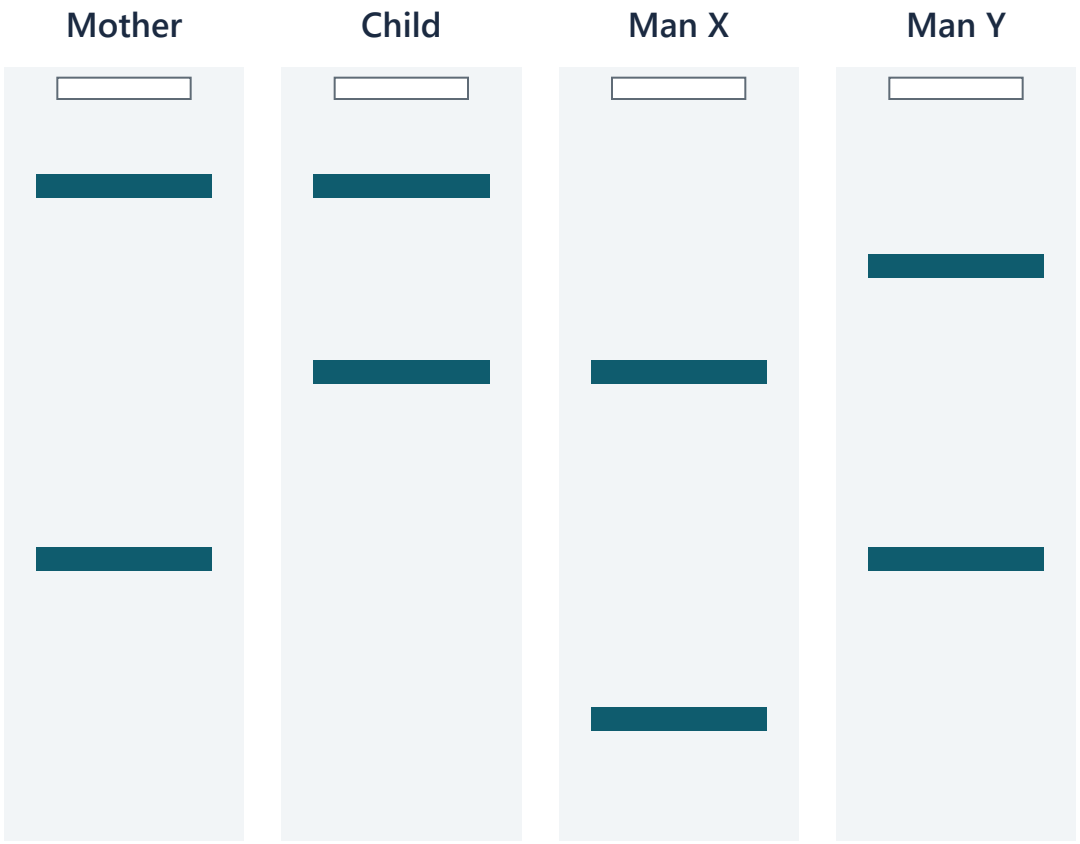
A profile from the crime scene is compared with each suspect's. **All bands must match** for the DNA to be from the same person.

Common mistake

A match shows DNA was at the scene, not how or when it got there. It is strong evidence, not proof of guilt.

Paternity: half from each parent

D1.1.5



Every band in the child comes from the **mother** or the **father**.

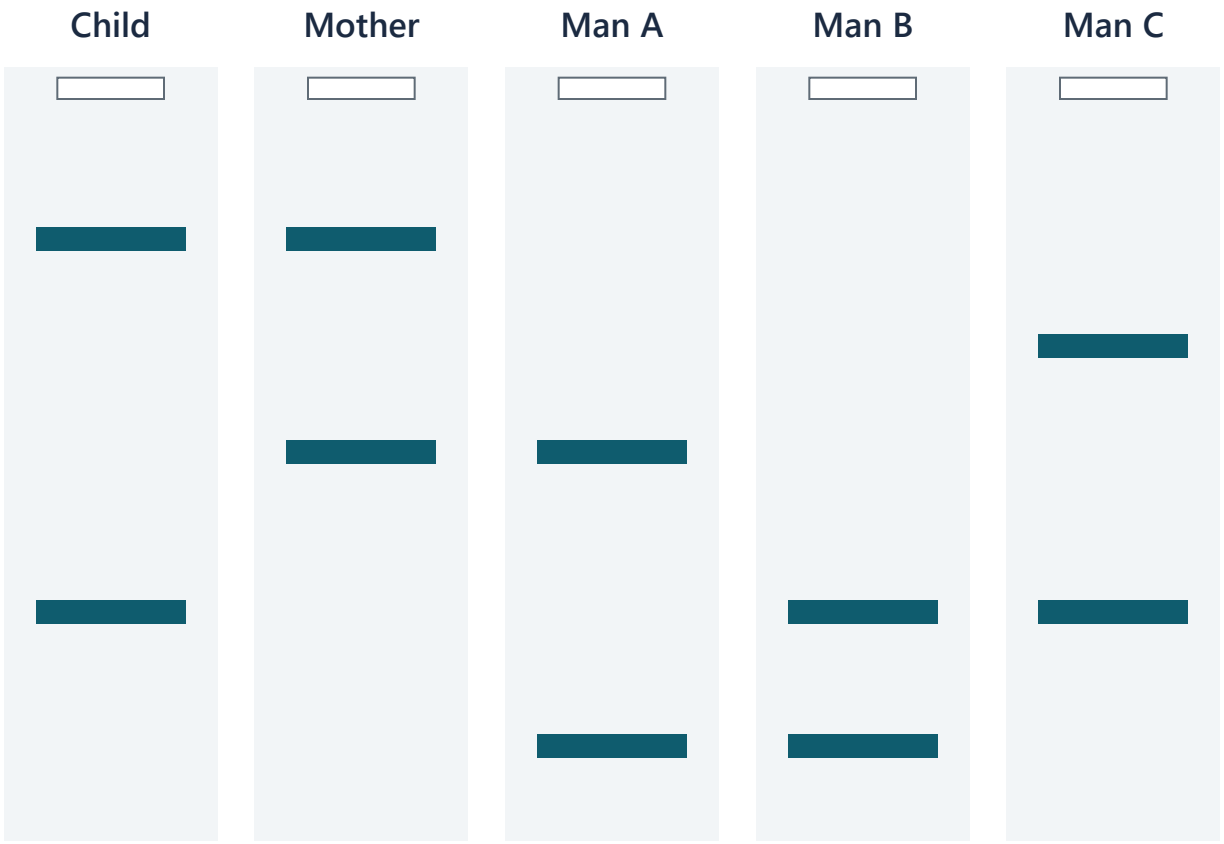
Cross off the child's bands that the mother explains. The father must have the rest.

Worked example

Child's top band matches the mother. The other band is not hers, so it came from the father. Man X has it; man Y does not, so man Y is excluded.

Practice: read the profiles

D1.1.5



1. Which band did the child get from the mother?
2. Which men could be the father?
3. Explain why more loci would be tested.
4. Explain why a DNA match is not proof of guilt.

Working with data: real PCR

D1.1.4–5

Copies of a target sequence starting from 10 molecules. The values are invented for practice; the pattern is realistic.

Cycles	Copies
0	10
5	320
10	10 240
20	about 9 000 000
35	about 1 000 000 000

1. Calculate the expected copies after 20 cycles.
2. Compare with the measured value.
3. Explain why copying slows in later cycles.
4. Each cycle takes 3 minutes: time for 35 cycles?

Check-in: class 3

1. Give two uses of PCR and gel electrophoresis.
2. Explain why DNA profiles differ between people.
3. State the rule for a match in a forensic profile.
4. Explain how a man can be excluded in a paternity test.
5. Explain why a DNA match is not proof of guilt.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
DNA polymerase unwinds the helix.	Helicase unwinds it; polymerase builds new strands.
In PCR, heating breaks covalent bonds.	Heating breaks hydrogen bonds between bases.
Larger fragments travel further in a gel.	Smaller fragments travel further.
Semi-conservative means half of each strand is new.	Each new molecule has one whole old and one whole new strand.

Summary: DNA replication

D1.1.1–5

	Key idea
D1.1.1	Copies needed for reproduction, growth and tissue replacement
D1.1.2	Semi-conservative; base pairing makes copies accurate
D1.1.3	Helicase separates; DNA polymerase builds
D1.1.4	PCR: 95, 55, 72 °C with Taq; gel separates by length
D1.1.5	STR profiles for forensics and paternity

Check that you can...

1. **Explain** why DNA is replicated and why replication is semi-conservative
2. **Outline** the roles of helicase and DNA polymerase
3. **Describe** PCR and the role of Taq polymerase
4. **Explain** how gel electrophoresis separates fragments
5. **Analyse** DNA profiles in forensic and paternity cases

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. A template strand reads A T T G C A. Write the complementary strand.

2. Name the enzyme that breaks hydrogen bonds between strands, and the enzyme that links nucleotides together.

3. Explain why fragments of DNA move through a gel, and why they separate.

Image credits

Gel electrophoresis 2.jpg – Mnolf – CC BY-SA 3.0
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DNA replication en.svg – LadyofHats Mariana Ruiz – Public domain
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